

Input / Output

Definition

Input/Output (I/O) refers to the process of taking input into a computer and producing output from it. Input devices allow users to give instructions to the computer, and output devices allow the computer to show the results.

1. Basic Idea

Main Concept

Every computer has physical devices for:

- acquiring **input**,
- producing **output**.

Without input and output devices, a computer would not be useful because users could neither give commands nor receive results.

2. Examples of I/O Devices

Common Devices

Many kinds of input and output devices exist. Some common examples are:

- **Keyboard** — input device
- **Monitor** — output device
- **Printer** — output device

More Examples

Other I/O devices may include:

- mouse,
- scanner,
- speakers,
- microphone,
- touchscreen.

3. Role of the Operating System

OS Responsibility

It is the responsibility of the **operating system** to manage all input and output devices.

Why OS Manages I/O

The operating system controls I/O devices so that:

- users can interact with the computer easily,
- programs can use devices without dealing with hardware details,
- the system works efficiently and correctly.

4. I/O Subsystem

Definition

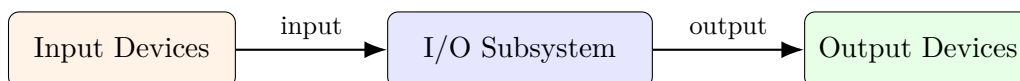
Every operating system contains an **I/O subsystem**. This subsystem is responsible for managing all I/O operations and devices.

Main Function

The I/O subsystem acts as a bridge between:

- the **hardware devices**, and
- the **operating system / user programs**.

4.1 Simple I/O Diagram



5. Device-Independent I/O Software

Meaning

Some parts of I/O software are **device independent**. This means they are general and can work with many or all I/O devices.

Importance

Device-independent software helps the operating system provide a common method of handling different devices, instead of writing everything separately for each one.

6. Device Drivers

Definition

A **device driver** is a special part of the I/O software that is designed for a specific hardware device.

Key Idea

Unlike device-independent software, device drivers are:

- device specific,
- written for particular hardware,
- responsible for controlling that device directly.

Examples

Different devices need different drivers, for example:

- keyboard driver,
- printer driver,
- display driver.

7. Device-Independent Software vs Device Drivers

Feature	Device-Independent I/O Software	Device Drivers
Nature	General purpose	Device specific
Use	Works for many devices	Works for one particular device
Role	Provides common I/O handling	Controls hardware directly
Examples	General I/O services	Keyboard, printer, monitor drivers